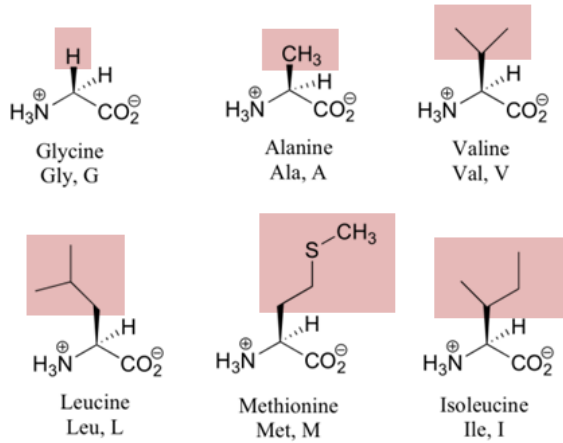


Bioinorganic Chemistry

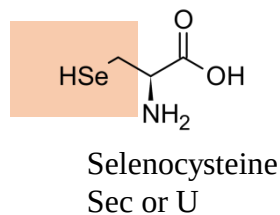
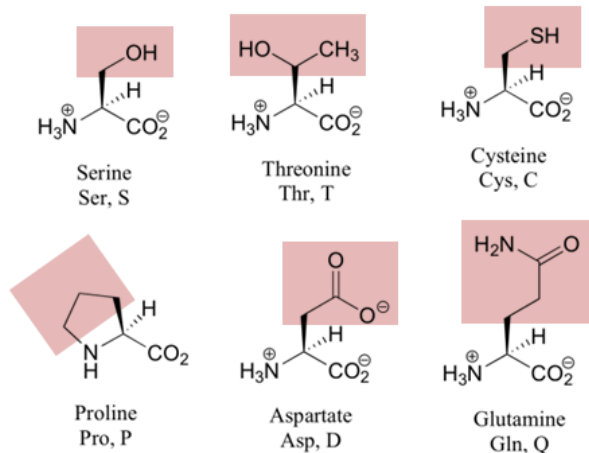
Amino Acid

Essential	Conditional/non-essential
Histidine	Alanine
Isoleucine	Arginine
Leucine	Aspartic acid
Lysine	Cysteine
Methionine	Glutamic acid
Phenylalanine	Glutamine
Threonine	Glycine
Tryptophan	Proline
Valine	Serine
	Tyrosine
	Asparagine
	Selenocysteine
	Pyrrolysine

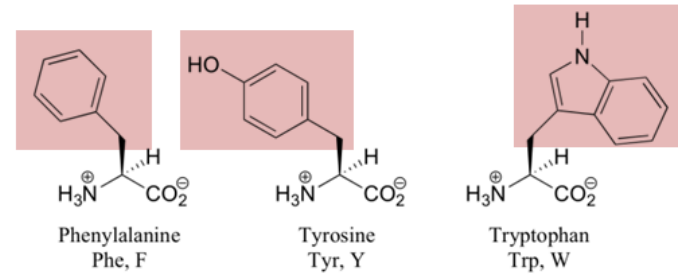
Nonpolar, aliphatic side groups



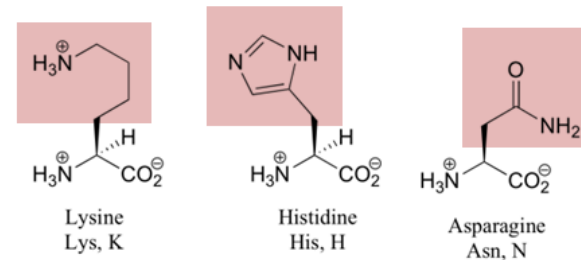
Polar, uncharged side groups



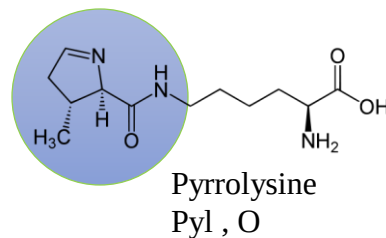
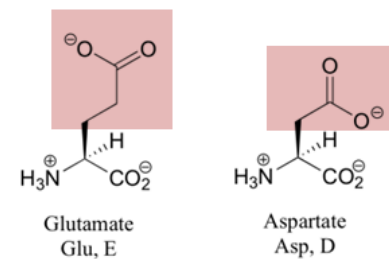
Aromatic side groups



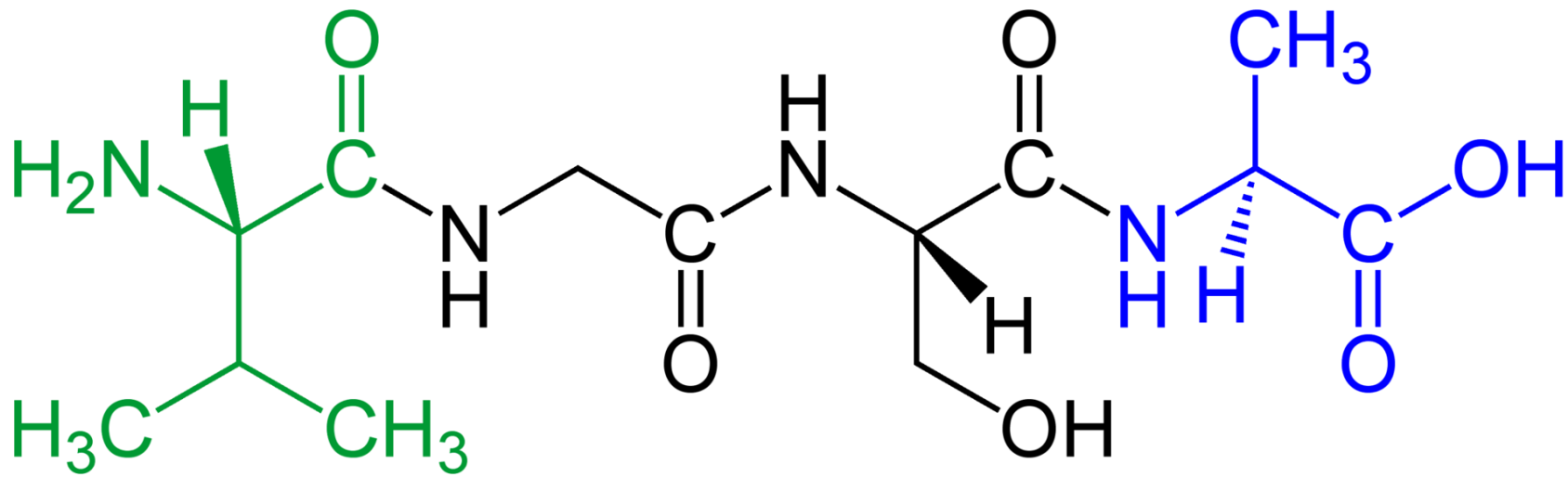
Positively charged side groups



Negatively charged side groups



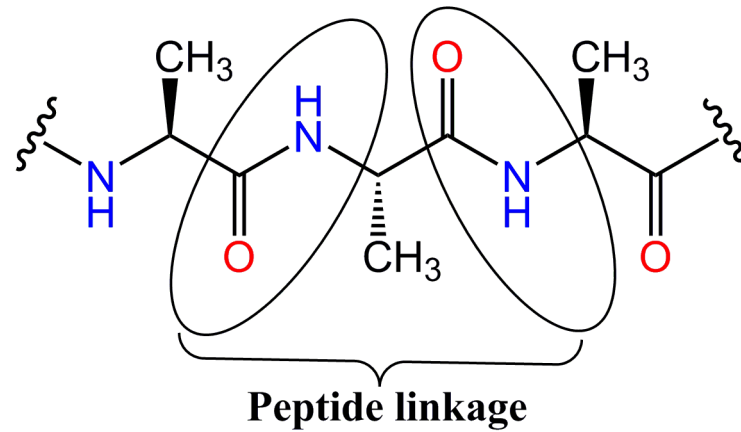
Amide Linkage



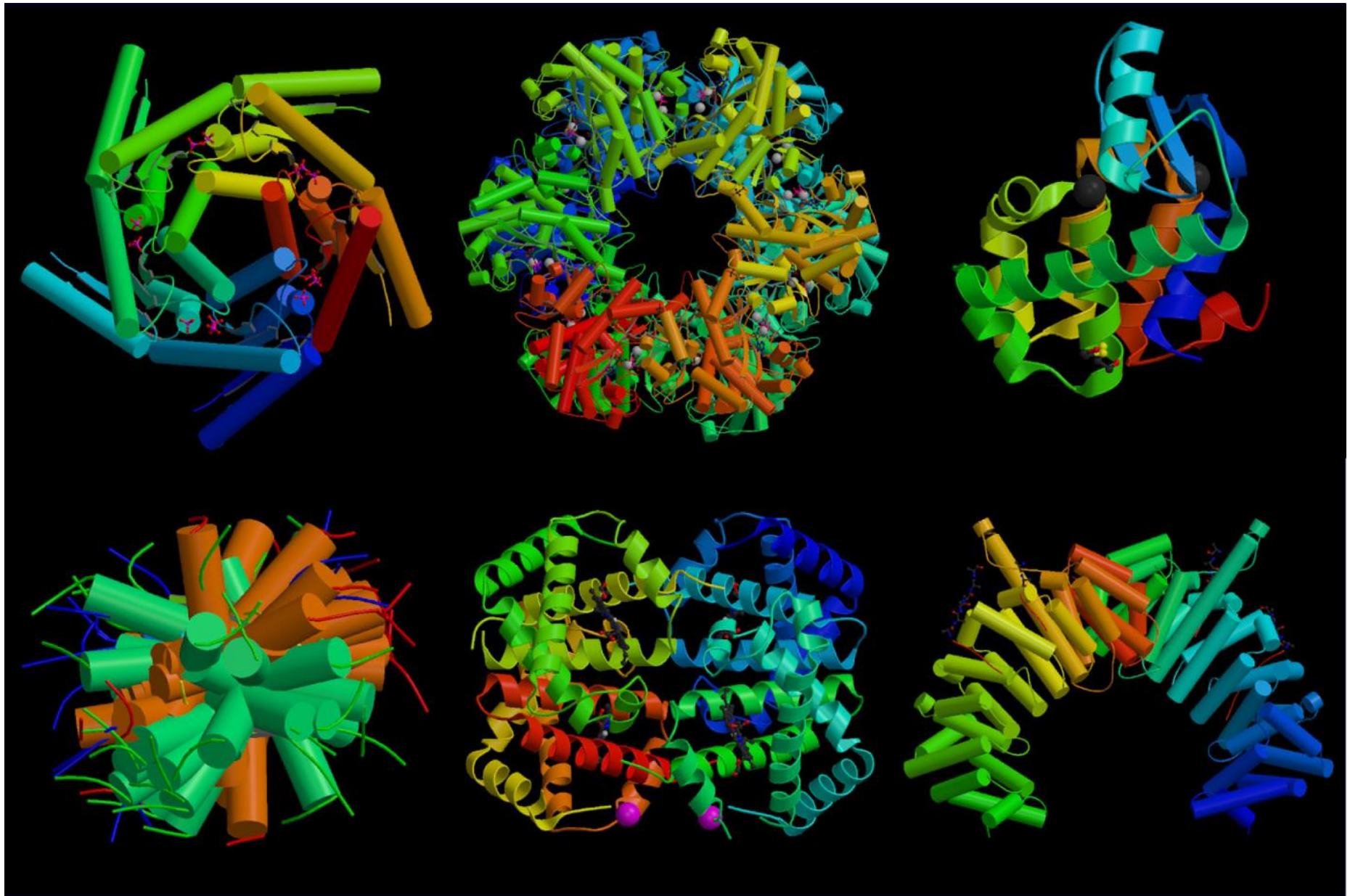
N-terminal

Val-Gly-Ser-Ala

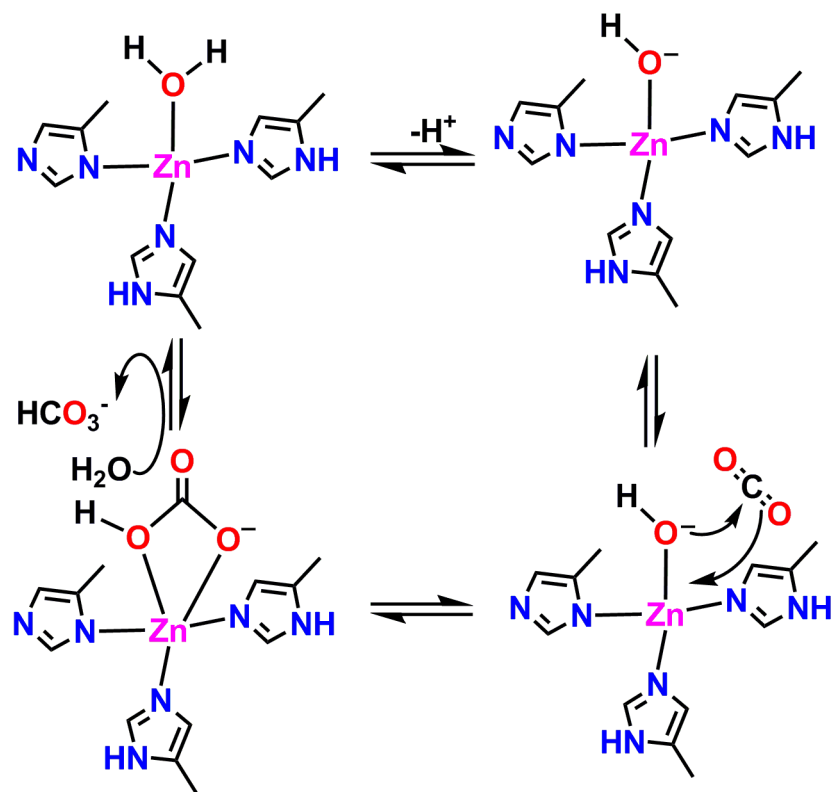
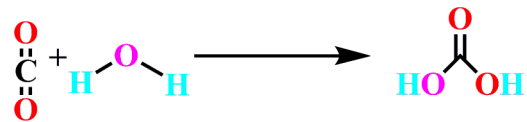
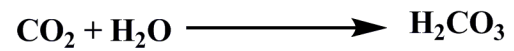
C-terminal



Proteins



<http://martin-protean.com/protein-structure.html>



Hemoglobin/Haemoglobin

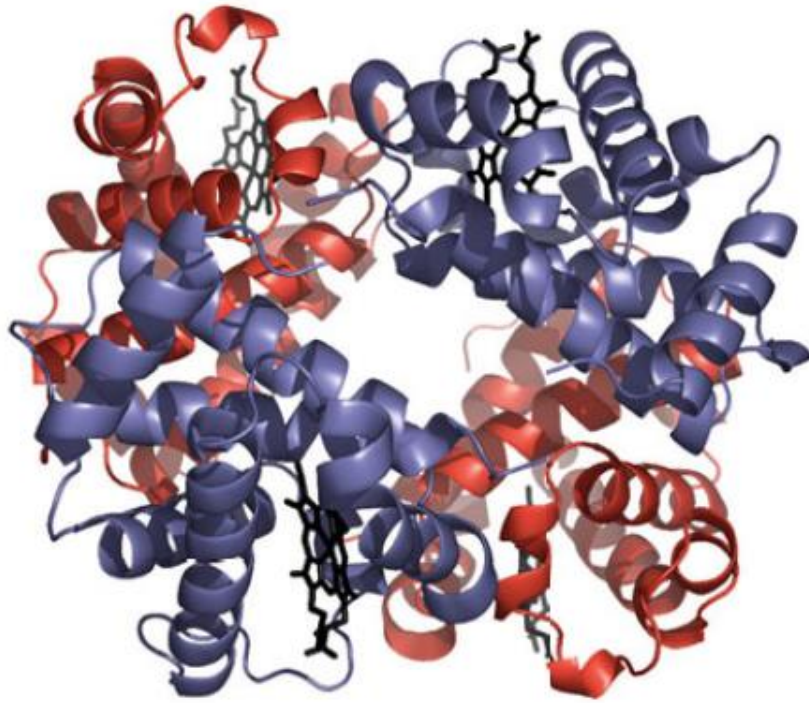
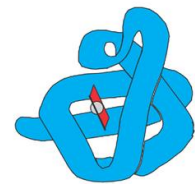
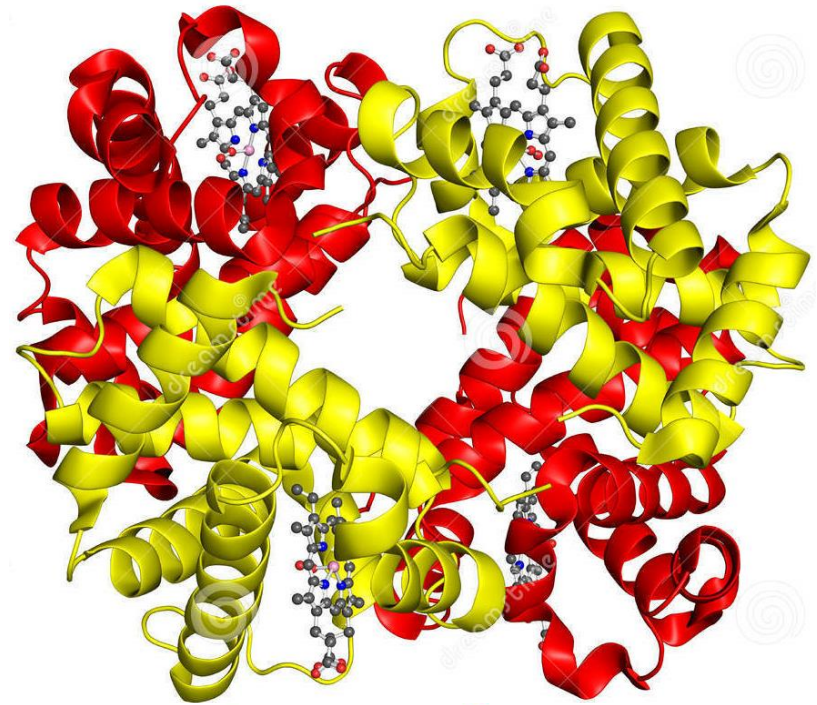
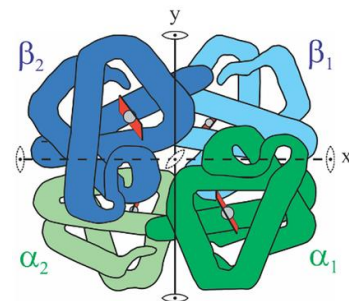


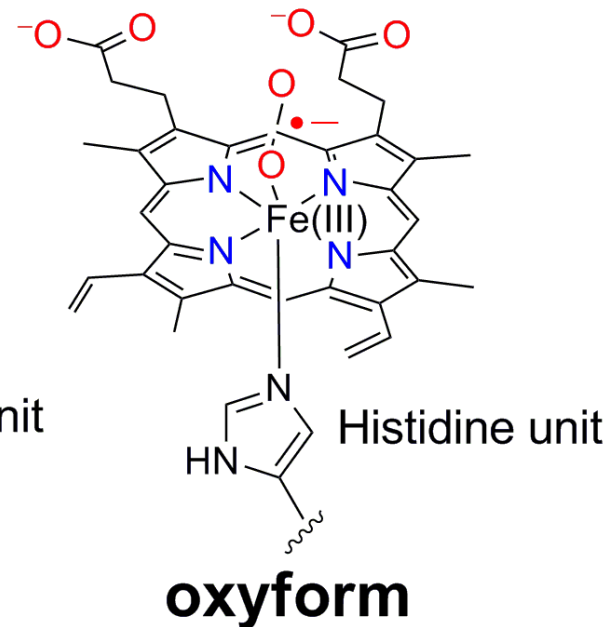
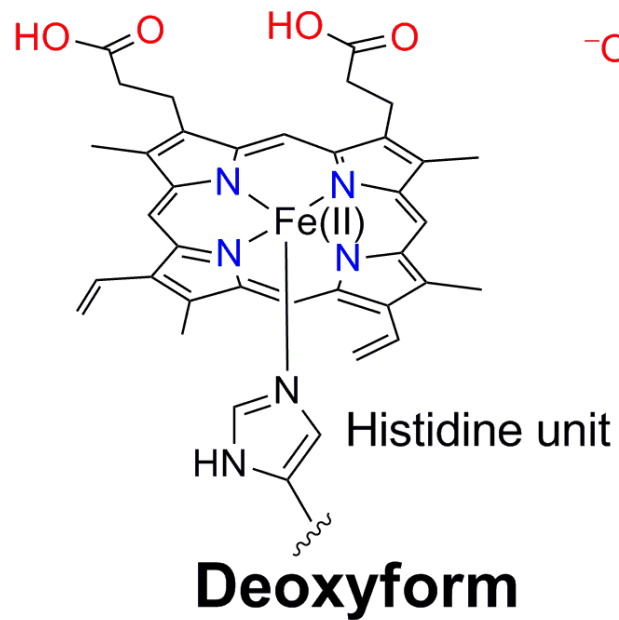
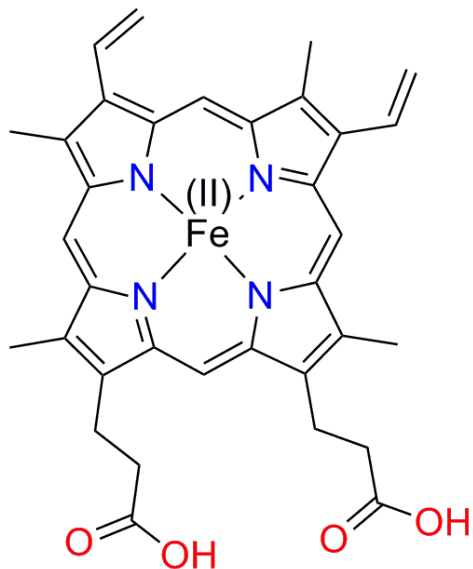
Figure 27.17 Haemoglobin is an $\alpha_2\beta_2$ tetramer. Its α and β subunits are very similar to myoglobin.



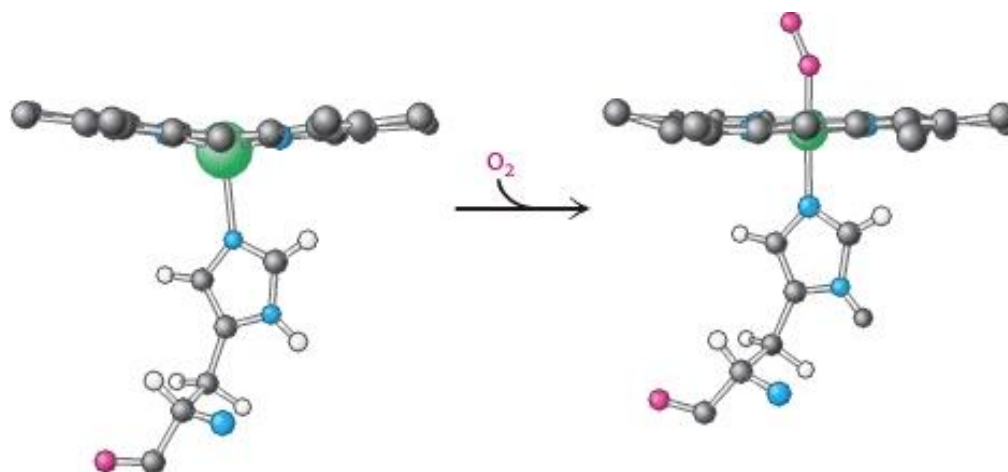
myoglobin



hemoglobin



Hemoglobin



Stability and Coordination Motif

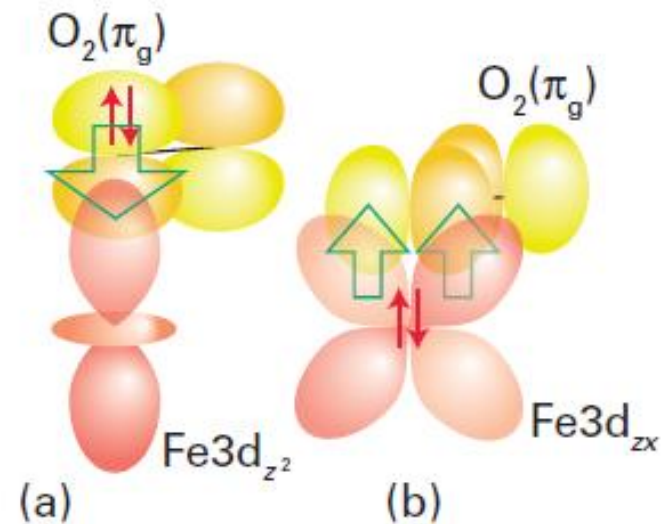
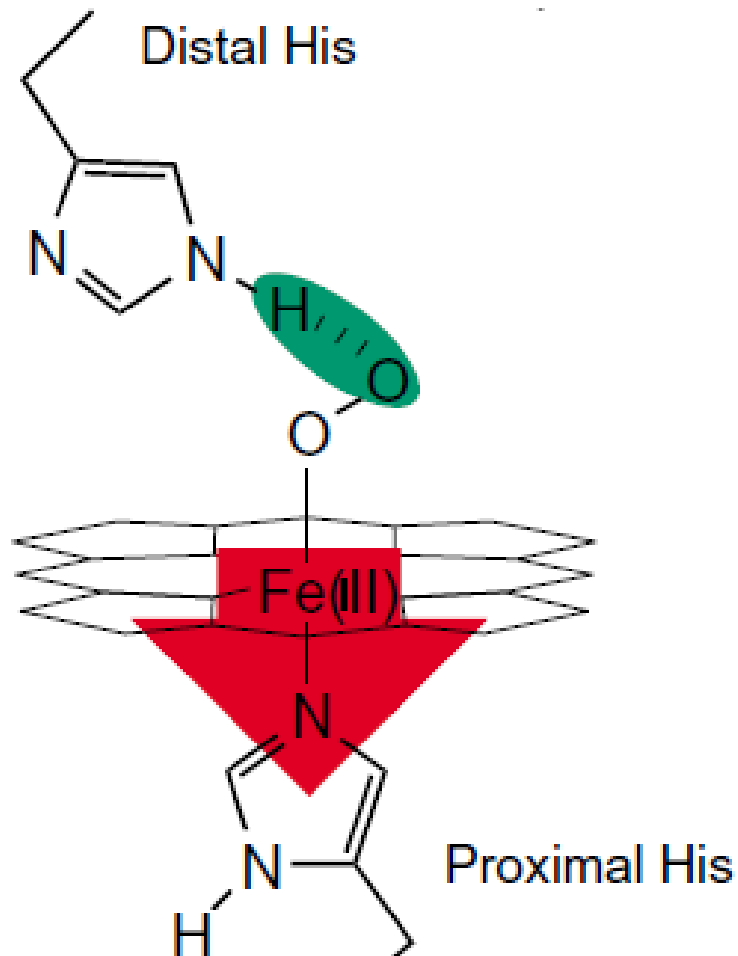


Figure 27.16 The orbitals used to form the Fe—O₂ adduct of myoglobin and haemoglobin. This model considers the O₂ ligand to be in a singlet state, in which the full $2\pi_g$ orbital donates an electron pair and the other $2\pi_g$ orbital acts as a p-electron pair acceptor.

Bohr Effect

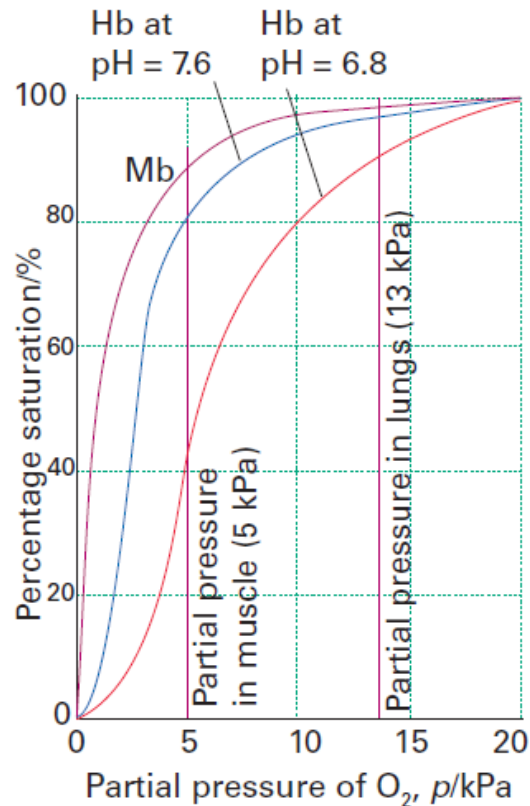


Figure 27.18 Oxygen binding curves for myoglobin and haemoglobin showing how cooperativity between the four sites in haemoglobin gives rise to a sigmoidal curve. The binding of the first O_2 molecule to haemoglobin is unfavourable, but it results in a greatly enhanced affinity for subsequent O_2 molecules.

